

Standard Operating Procedure & Troubleshooting Manual

Equipment: Strand Pelletiser Line PL-202 (water bath, air knife, pelletiser)

Document Ref: SOP-ENG-PL202-REV2

Plant Location: Polymer Compounding Facility (Line 2, after extruder EX-204)

Applicability: Process Operators, Maintenance Technicians, Shift Supervisors

1. Equipment Description

Line **PL-202** turns the 24 melt strands from extruder EX-204 (PP + 20 % CaCO₃) into pellets. Strands are cooled in a 4 m water bath, dried by an air knife, then pulled by feed rolls into a rotating cutter that cuts them against a fixed bed knife. Pellets pass through a centrifugal dryer and a classifier screen that removes fines and oversize pellets. Specified pellet length is 3.0–3.5 mm.

2. Operating Envelope & Alarm Limits

Table 1: Operating envelope and alarm limits

Parameter	Normal Range	Warning	Critical Alarm	Trip / Action
Water bath temperature	30 – 40 °C	below 25 °C or above 45 °C	below 20 °C or above 50 °C	Supervisor review
Strands running	24 of 24	22 – 23	below 20	Rethread
Pellet length	3.0 – 3.5 mm	below 2.8 mm or above 3.8 mm	below 2.5 mm or above 4.5 mm	Quality hold
Pellet moisture after dryer	below 0.10 %	0.10 %	0.20 %	Quality hold
Fines (dust) in product	below 0.5 %	0.5 %	1.0 %	Quality hold
Air knife pressure	3 – 5 bar	below 2.5 bar	below 2.0 bar	Check air supply

Pellet length, moisture and fines are checked by the line technician every hour.

3. Fault Matrix

Table 2: Faults, signs, causes and actions

Fault	Typical Signs	Likely Causes	Operator Action
FAULT-A: Strand breaks	• Strands snap in the bath or at the feed rolls • Strand count drops	Bath too cold (brittle strands), blocked die holes or contamination, unstable extruder output, pull speed too high.	1. Check bath temperature. 2. Check die and extruder melt pressure. 3. Rethread under lockout. 4. Adjust pull speed.
FAULT-B: Long or irregular pellets	• Length out of spec • 'Tails' or pellets joined in chains	Dull rotor or bed knife, wrong knife gap, cutter speed not matched to strand speed, strands too soft (bath too warm).	1. Check cutter-to-strand speed ratio. 2. Check bath temperature. 3. Request knife change and gap setting.
FAULT-C: Wet pellets	• Moisture above 0.10 % • Water in product bags	Air knife pressure low or nozzles blocked, centrifugal dryer screen blocked, bath level too high.	1. Check air knife pressure and nozzles. 2. Clean dryer screen. 3. Quality hold on affected product.

FAULT-D: Fines and dust	• Fines above 0.5 % • Dusty bags	Strands too cold (brittle cut), dull knives, cutter speed too high.	1. Raise bath temperature within range. 2. Check knives. 3. Check classifier screen.
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4. Operating Rules

- Lock out the cutter before rethreading or knife work. The rotor keeps turning after stop; wait until it has stopped completely.
- Never put hands near the feed rolls while the line is running.
- Quality-hold all product made since the last good check when a fault is found.
- Settings outside the operating envelope need shift supervisor approval.